MedGemma (4b-it) for CURE-Bench

Internal Model Reasoning

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NEURAL INFORMATION PROCESSING SYSTEMS

Evaluation of MedGemma on CURE-Bench

Various open-source models fine-tuned on medical reasoning and decision-making were preliminarily tested and evaluated on the Validation Set

- dousery/medical-reasoning-gpt-oss-20b
- BioMistral/BioMistral-7B
- google/medgemma-4b-it
- google/medgemma-27b-it

Medgemma-4b-it was chosen for the Test Set based on its superior performance on the Validation Set, with a smaller model size, as well parseability of its output responses as per required format.

Parameters

- Max New Tokens = 256
- Do Sample = False
  - => Greedy Search
  - => No random sampling, Deterministic Output for repeatability

Prompt

Role Instruction

You are an expert doctor.

System Instruction

SYSTEM INSTRUCTION: {role_instruction} Reply strictly as per schema
{{ 'answer': '<letter of best option>',
'reasoning': '<reasoning explaining why>'
}}

RunTime

Hardware	T4 GPU
Language	Python 3.11
Libraries	Transformers 4.56.0

Results

Validation Set

Model scores on the Validation Set

Model	Accuracy	Macro-F1
Gemma-4b-it Run 1	0.77	0.7
Gemma-4b-it Run 2	0.7	0.65
BioMistral	0.64	0.52
Gemma-4b-it + BioMistral (Any)	0.84	

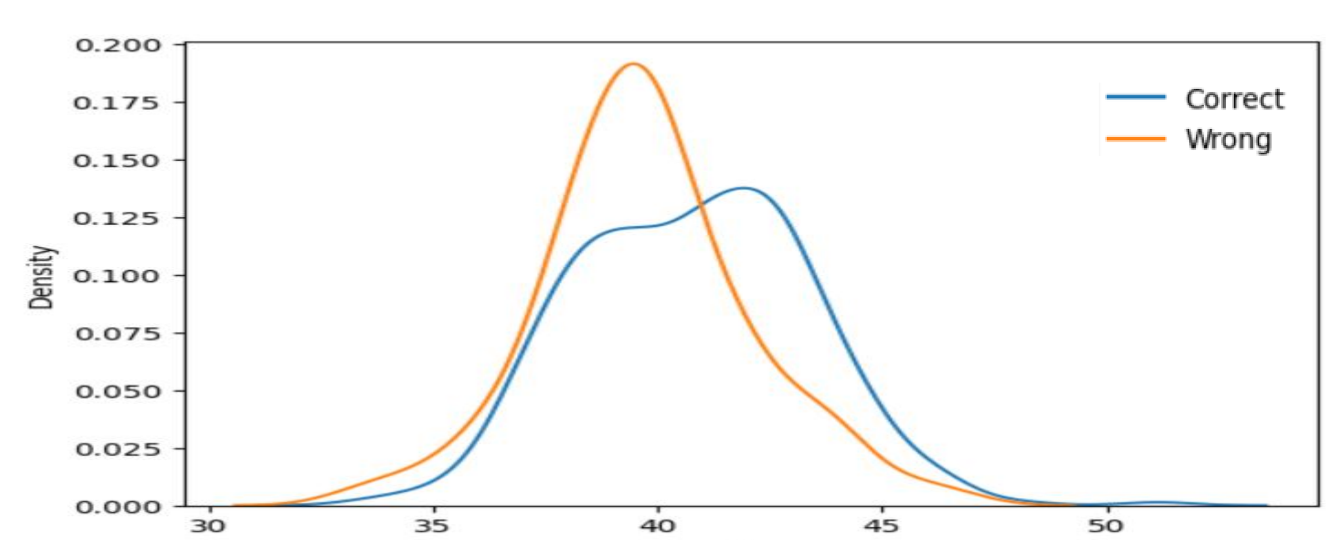
Best results were obtained when considering the Top 2 Accuracy taking into account the output results of Gemma-4b and BioMistral.

Score was not generated for Medgemma-27b-it and Medical-Reasoning GPT OSS as several of the outputs did not adhere to specified format.

Confidence Scores

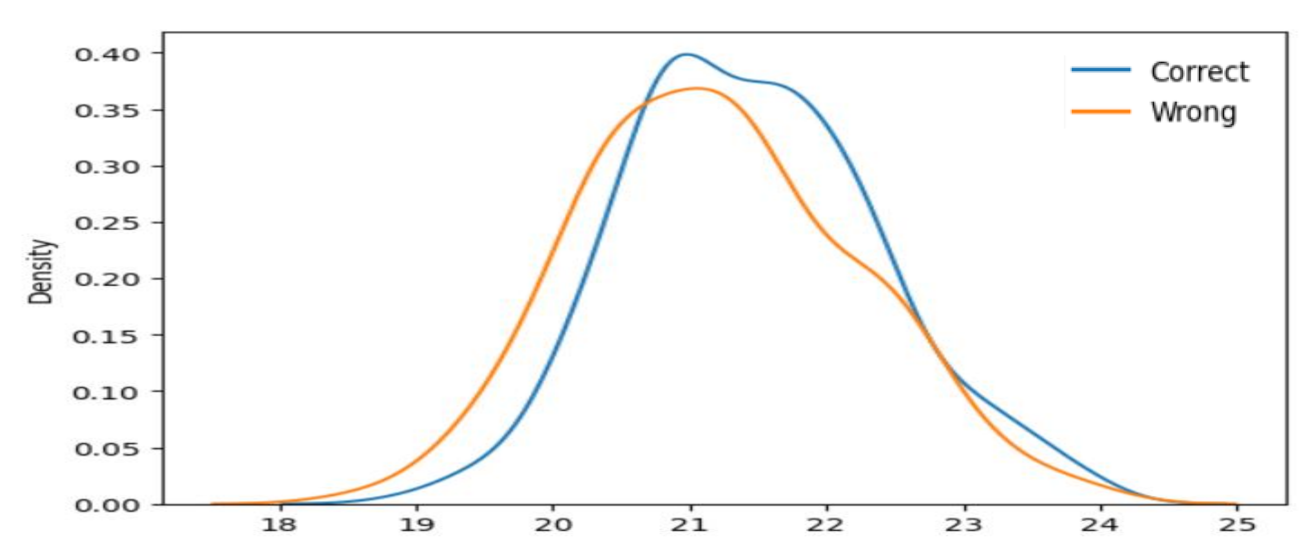
A proxy Confidence Score was calculated for the generated outputs by taking the Average of Maximum Logit (Unnormalized Log Probability) at each token prediction step.

Distribution of MedGemma Confidence Scores Across Different Answers




```
F_onewayResult(statistic=13.693488993397034, pvalue=0.00024160525112321355)
```

Distribution of BioMistral Confidence Scores Across Different Answers




```
F_onewayResult(statistic=7.842694351914448, pvalue=0.005320896119355765)
```

Retrieval Augmented Generation

NER from Question and Options

Named Entity Recognition (NER)

SpaCy, ClinicalBERT, BioBERT and various models were evaluated qualitatively in their ability to recognize drug names. tner/xlm-roberta-base-bc5cdr performs best.

Extract Matches from FDA DB

FDALabel Database

Documents having matches with drug names were selected

Extract passages with highest cosine similarity to question and options from FDA DB

Similarity Matching

Passages from the selected documents having highest cosine similarity to question and options are extracted

Pass extracted passges to model as Additional Information

Additional Information to Model

Relevant passages are passed to the model as additional information

References

MedGemma - Google DeepMind
[2402.10373] BioMistral: A Collection of Open-Source Pretrained Large Language Models for Medical Domains
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[2005.11401] Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks
FDALabel: Full-Text Search of Drug Product Labeling | FDA